Mohimenul Kabir (Mahi)

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Google Scholar: See Google Scholar

Research interests

Formal Methods, Logic Programming, Answer Set Programming, Artificial Intelligence, Metaheuristic

Education

2020 – **Present** **National University of Singapore** (NUS) – Singapore
PhD in Computer Science
Advisor: Kuldeep S. Meel *GPA: 4.42* (out of 5.00).

2013 – 2017 **Bangladesh University of Engineering and Technology** (BUET) – Bangladesh
BSc in Computer Science
Mentors: M Sohel Rahman *GPA: 3.77* (out of 4.00).

Honors and scholarships

2023 NUS SoC Research Incentive Award

2015 University Merit Scholarship (Bangladesh University of Engineering and Technology)

2014 University Merit Scholarship (Bangladesh University of Engineering and Technology)

Publications

2024 **On Lower Bounding Minimal Model Count**
Mohimenul Kabir and Kuldeep S Meel
Theory and Practice of Logic Programming (Special issue of ICLP 2024).

2024 **Exact Answer Set Counting with Compact Encoding**
Mohimenul Kabir, Supratik Chakraborty and Kuldeep S Meel
*Proceedings of the AAAI Conference on Artificial Intelligence 2024 (**Invited in Model Counting and Sampling Workshop 2024**).*

2023 **A Fast and Accurate ASP Counting Based Network Reliability Estimator**
Mohimenul Kabir and Kuldeep S Meel
International Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR) 2023. (**Invited in Model Counting Competition Workshop 2023**)

2023 **Multi-objective optimization and heuristic based solutions for evacuation modeling**
Mohimenul Kabir, Jaiaid Mobin, Muhammad Ali Nayeem, Muhammad Ahsanul Habib, and M. Sohel Rahman
Transportation Research Interdisciplinary Perspectives.

2022 **ApproxASP – a scalable approximate answer set counter**
Mohimenul Kabir, Flavio O Everardo, Ankit K Shukla, Markus Hecher, Johannes Klaus Fichte, and Kuldeep S Meel
Proceedings of the AAAI Conference on Artificial Intelligence 2022. (**Invited in ICLP 2022 and Indian SAT-SMT school 2022**)

2020 **Evaluation of classification and forecasting methods on time series gene expression data**
Nafis Irtiza Tripto, **Mohimenul Kabir**, Md Shamsuzzoha Bayzid, Atif Rahman
PLOS ONE.

2020 **ADACT: a tool for analysing (dis) similarity among nucleotide and protein sequences using minimal and relative absent words**
Mujtahid Akon, Muntashir Akon, **Mohimenul Kabir**, M Saifur Rahman, M Sohel Rahman
Bioinformatics.

2018 **A simple, fast, filter-based algorithm for circular sequence comparison**
Md Azim, Aashikur Rahman, **Mohimenul Kabir**, M Sohel Rahman
In International Workshop on Algorithms and Computation (WALCOM).

Future Research Goals

July 2024 – Present **Minimal Model Counting**
Mentor: Kuldeep S Meel
We aim to engineer a minimal model counter for the Boolean formula. We are also seeking real-world applications of minimal model counting.

January 2024 – Present **Answer Set Counting in Bioinformatics**
Mentor: Kuldeep S Meel
We are applying answer set counters to real-world system biology problems. Our purpose is to perform inference on bioinformatic problems.

Research Visits

October 2024 **Texas Tech University**

January – June 2024 **University of Toronto**

Teaching experience

Fall 2023 **Teaching assistant, IT5005: Introduction to Artificial Intelligence (NUS)**

Spring 2021 & Spring 2023 **Teaching assistant, CS4244: Knowledge Representation and Reasoning (NUS)**

Fall 2021 **Teaching assistant, CS3243: Introduction to Artificial Intelligence (NUS)**

Spring 2022 **Teaching assistant, CS4246/CS5446: AI Planning and Decision Making (NUS)**

Fall 2022 **Teaching assistant, CS2040: Data Structures and Algorithms (NUS)**

Talks and tutorials

August 2024 — Exact ASP Counting with Compact Encodings
Model Counting and Sampling workshop 2024

February 2024 — Exact ASP Counting with Compact Encodings
AAAI 2024

June 2023 — A Fast and Accurate ASP Counting Based Network Reliability Estimator
LPAR 2023

December 2022 — ApproxASP – a scalable approximate answer set counter
SAT SMT Winter School 2022

February 2022 — ApproxASP – a scalable approximate answer set counter
AAAI 2022

March 2018 — A simple, fast, filter-based algorithm for circular sequence comparison
WALCOM 2018

Mentorship and service

May 2022 – July 2022 — **National University of Singapore (Academic Advisor)**
My responsibilities: I mentored two undergraduate students on a research project. The title of the project: "Approximate Answer Set Sampling"

Technical skills

Programming languages
Proficient in: C, C++, Java, Python, R

Software
LaTeX, Git, VirtualBox

Languages
English (fluent)